

VIT Bhopal University

Project Report

Umesh Restaurant Management System

Python console program for menu and billing

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Course: B.Tech CSE Core, First Year

College: VIT Bhopal

GitHub: github.com/ukhandare30-commits/Restaurant-Management

1. Introduction

I made a Python program for a restaurant called Umesh Restaurant. It runs in the terminal. The owner can change the menu and the customer can order food and get a bill. GST is 5%.

I used functions, lists, files, loops and if-else. Menu is saved in text files. There is no extra library.

2. Problem Statement

If a restaurant writes menu and bills on paper, prices get messy and GST can be wrong. I wanted a small program where the owner updates dishes and the customer gets a proper bill.

3. Functional Requirements

There are 3 parts.

Owner part: password is admin123. Owner can add, change, delete and see items in starters, main course, desserts, drinks and breads.

Customer part: customer opens a menu, picks items and quantity, can edit or remove items.

Bill part: after confirm, program prints subtotal, GST 5% and grand total.

4. Non-functional Requirements

Point	What I did
Easy to use	Number menus and short messages
Password	Owner needs admin123
Errors	If price or number is wrong, program says invalid
Data stays	Menu is saved in .txt files

5. System Architecture

User uses main.py. Owner functions change the text files. Customer functions read the files and make a bill. Database.py is only used once to create the first menu.

File	Use
main.py	Whole program
Database.py	Makes first menu files
starters.txt etc.	Saved dishes
README.md	How to run
statement.md	Problem and scope

6. Design Diagrams

What each person can do:

Owner	Customer
Enter password	See menu
Add / change / delete dish	Add item to order

See menu	Edit or remove item
	Confirm and get bill

How the program runs:

Start -> main menu -> Owner or Customer or Exit.

Owner -> password. If admin123 then add/edit/delete/view. If wrong, it asks again.

Customer -> pick category -> order items -> confirm -> bill with GST.

Bill steps: customer chooses food -> program reads file -> customer enters number and qty -> program keeps order list -> confirm -> print bill.

Storage is not SQL. One line in file looks like: Paneer Tikka|Starter|220

7. Design Decisions

I kept it as a console app because that is what we did in class. I used text files instead of a database so I can explain it. Order is a list. GST is 0.05 in the code. Password is just a string check so the teacher can test owner menu.

8. Implementation

Function	What it does
read_menu()	Reads file into a list
view_menu()	Prints menu
add_item / modify_item / delete_item	Owner changes file
order_item / edit_order / remove_order_item	Customer order
generate_bill()	Subtotal + GST + total
owner_menu / customer_menu / main	Menus

If password is admin123, owner menu opens. Else it says incorrect password.

9. Screenshots / Results

Screenshots are from my GitHub files main.py and Database.py. Typed answers are written after each Enter line. Confirm order question is: Do you want to confirm your order? (y/n):

Main menu:

```

    UMESH RESTAURANT
    1. Owner
    2. Customer
    3. Exit
    Enter your choice: 2

```

Owner section

I opened Owner, typed a wrong password, then admin123.

```
UMESH RESTAURANT

1. Owner
2. Customer
3. Exit
Enter your choice: 1
Enter Password: hello
Incorrect Password try again
Enter Password: admin123

OWNER MENU
1. Add Item
2. Modify Item
3. Delete Item
4. View Menu
5. Exit
Enter your choice: 4
```

Owner adding an item to Starters:

```
OWNER MENU
1. Add Item
2. Modify Item
3. Delete Item
4. View Menu
5. Exit
Enter your choice: 1

Select Menu
1. Starters
2. Main Course
3. Desserts
4. Drinks
5. Breads
Enter your choice: 1

ADD ITEM - STARTERS
Enter dish name: Corn Cheese Balls
Enter food type: Starter
Enter price: 200
Item added successfully.
```

Owner viewing menus (from Database.py / txt files):

Starters menu (items and prices):

Select Menu
1. Starters
2. Main Course
3. Desserts
4. Drinks
5. Breads
Enter your choice: 1

UMESH RESTAURANT
MENU CARD

STARTERS

NO.	DISH NAME	TYPE	PRICE
1	Tomato Basil Soup	Starter	Rs. 150
2	Paneer Tikka	Starter	Rs. 220
3	Crispy Corn	Starter	Rs. 180
4	Veg Spring Rolls	Starter	Rs. 200
5	Hara Bhara Kabab	Starter	Rs. 190
6	French Fries	Starter	Rs. 160
7	Cheese Balls	Starter	Rs. 210
8	Chilli Paneer	Starter	Rs. 240
9	Garlic Bread	Starter	Rs. 170
10	Peri Peri Fries	Starter	Rs. 190
11	Stuffed Mushrooms	Starter	Rs. 230
12	Veg Seekh Kebab	Starter	Rs. 220
13	Onion Rings	Starter	Rs. 180
14	Paneer Pakora	Starter	Rs. 200
15	Aloo Tikki	Starter	Rs. 160
16	Dahi Ke Kebab	Starter	Rs. 220
17	Peanut Masala	Starter	Rs. 140
18	Tandoori Momos	Starter	Rs. 210
19	Cheese Nachos	Starter	Rs. 230
20	Veg Cutlet	Starter	Rs. 170

Main course menu:

Select Menu
1. Starters
2. Main Course
3. Desserts
4. Drinks
5. Breads
Enter your choice: 2

UMESH RESTAURANT
MENU CARD

MAIN COURSE

NO.	DISH NAME	TYPE	PRICE
1	Margherita Pizza	Main Course	Rs.350
2	Paneer Butter Masala	Main Course	Rs.280
3	Veg Biryani	Main Course	Rs.250
4	White Sauce Pasta	Main Course	Rs.300
5	Veg Manchurian	Main Course	Rs.270
6	Veg Fried Rice	Main Course	Rs.240
7	Paneer Tikka Masala	Main Course	Rs.320
8	Dal Makhani	Main Course	Rs.260
9	Shahi Paneer	Main Course	Rs.300
10	Mushroom Masala	Main Course	Rs.280
11	Veg Hakka Noodles	Main Course	Rs.230
12	Farmhouse Pizza	Main Course	Rs.400
13	Arrabbiata Pasta	Main Course	Rs.290
14	Malai Kofta	Main Course	Rs.310
15	Kadai Paneer	Main Course	Rs.300
16	Palak Paneer	Main Course	Rs.280
17	Chole Bhature	Main Course	Rs.250
18	Rajma Chawal	Main Course	Rs.220
19	Veg Lasagna	Main Course	Rs.320
20	Stuffed Capsicum	Main Course	Rs.290

Desserts menu:

Select Menu
1. Starters
2. Main Course
3. Desserts
4. Drinks
5. Breads
Enter your choice: 3

UMESH RESTAURANT
MENU CARD

DESSERTS

NO.	DISH NAME	TYPE	PRICE
1	Gulab Jamun	Dessert	Rs.100
2	Brownie	Dessert	Rs.150
3	Ice Cream	Dessert	Rs.120
4	Cheesecake	Dessert	Rs.180
5	Chocolate Mousse	Dessert	Rs.160
6	Fruit Custard	Dessert	Rs.130
7	Chocolate Lava Cake	Dessert	Rs.190
8	Rasmalai	Dessert	Rs.140
9	Kulfi	Dessert	Rs.110
10	Apple Pie	Dessert	Rs.170
11	Caramel Pudding	Dessert	Rs.150
12	Sizzling Brownie	Dessert	Rs.220
13	Rasgulla	Dessert	Rs.120
14	Jalebi	Dessert	Rs.100
15	Tiramisu	Dessert	Rs.200
16	Red Velvet Cake	Dessert	Rs.190
17	Chocolate Donut	Dessert	Rs.130
18	Kesar Pista Kulfi	Dessert	Rs.140
19	Shahi Tukda	Dessert	Rs.160
20	Mango Cheesecake	Dessert	Rs.200

Drinks menu:

Select Menu
1. Starters
2. Main Course
3. Desserts
4. Drinks
5. Breads
Enter your choice: 4

UMESH RESTAURANT
MENU CARD

DRINKS

NO.	DISH NAME	TYPE	PRICE
1	Fresh Lime Soda	Drink	Rs.100
2	Cold Coffee	Drink	Rs.150
3	Mango Shake	Drink	Rs.160
4	Coca Cola	Drink	Rs.80
5	Masala Chaas	Drink	Rs.90
6	Chocolate Shake	Drink	Rs.170
7	Oreo Shake	Drink	Rs.190
8	Strawberry Shake	Drink	Rs.160
9	Virgin Mojito	Drink	Rs.140
10	Iced Tea	Drink	Rs.120
11	Lemonade	Drink	Rs.100
12	Watermelon Juice	Drink	Rs.130
13	Hot Coffee	Drink	Rs.90
14	Hot Chocolate	Drink	Rs.150
15	Pineapple Juice	Drink	Rs.140
16	Orange Juice	Drink	Rs.130
17	Blue Lagoon	Drink	Rs.160
18	Vanilla Shake	Drink	Rs.150
19	Green Apple Mojito	Drink	Rs.170
20	Cold Chocolate	Drink	Rs.160

Breads menu:

Select Menu
1. Starters
2. Main Course
3. Desserts
4. Drinks
5. Breads
Enter your choice: 5

UMESH RESTAURANT
MENU CARD

BREADS

NO.	DISH NAME	TYPE	PRICE
1	Butter Naan	Bread	Rs .60
2	Garlic Naan	Bread	Rs .80
3	Tandoori Roti	Bread	Rs .40
4	Cheese Naan	Bread	Rs .100
5	Lachha Paratha	Bread	Rs .90
6	Missi Roti	Bread	Rs .70
7	Plain Naan	Bread	Rs .50
8	Butter Roti	Bread	Rs .45
9	Stuffed Kulcha	Bread	Rs .110
10	Aloo Paratha	Bread	Rs .100
11	Paneer Paratha	Bread	Rs .130
12	Roomali Roti	Bread	Rs .60
13	Onion Kulcha	Bread	Rs .100
14	Amritsari Kulcha	Bread	Rs .120
15	Pudina Paratha	Bread	Rs .90
16	Methi Paratha	Bread	Rs .85
17	Cheese Garlic Naan	Bread	Rs .120
18	Tandoori Paratha	Bread	Rs .100
19	Peshawari Naan	Bread	Rs .130
20	Ajwain Roti	Bread	Rs .55

Customer side after opening starters:

WHICH MENU WOULD YOU LIKE TO SEE?

1. Starters
 2. Main Course
 3. Desserts
 4. Drinks
 5. Breads
 6. Confirm Order
- Enter your choice: 1

UMESH RESTAURANT
MENU CARD

STARTERS

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19	Cheese Nachos	Starter	Rs.230
20	Veg Cutlet	Starter	Rs.170

MENU OPTIONS

1. Order an Item
2. View Current Order
3. Edit Current Order
4. Remove an Item
5. Change Menu
6. Confirm Order

Enter your choice: 1

Enter item number: 1

Enter quantity: 2

Tomato Basil Soup added to your order.

Sample bill: 2 Tomato Basil Soup. Subtotal 300, GST 15, total 315.

```
MENU OPTIONS
1. Order an Item
2. View Current Order
3. Edit Current Order
4. Remove an Item
5. Change Menu
6. Confirm Order
Enter your choice: 6

CURRENT ORDER
-----
NO. ITEM                                QTY      TOTAL
-----
1  Tomato Basil Soup                    2        Rs.300
-----
Do you want to confirm your order? (y/n): y

                UMESH RESTAURANT
                  BILL
-----
ITEM                                QTY      TOTAL
-----
Tomato Basil Soup                    2        Rs.300
-----
Subtotal:                                Rs.300.0
GST (5%):                                Rs.15.0
Grand Total:                            Rs.315.0
-----
Thank you for visiting Umesh Restaurant!
```

10. Testing

Test	What I did	Result
Menu files	python Database.py	5 txt files created
Start	python main.py	Main menu came
Wrong password	typed hello	Incorrect password
Right password	typed admin123	Owner menu opened
Order	2 soup and confirm	Bill showed GST

11. Challenges

Making the menu look like a table with spaces was annoying. After delete I had to write the whole file again. I first uploaded files on the GitHub website so commits say Add files via upload. I also made two repos by mistake. The one with the hyphen is the final repo.

12. What I learned

I got better at functions and files. I understood how a small password can split owner and customer. I also learned how to put README and statement.md on GitHub.

13. Future work

Later I can save bills, make a GUI or use a database. Not doing that in this submission.

14. References

Python docs: <https://docs.python.org/3/>

My repo: <https://github.com/ukhandare30-commits/Restaurant-Management>